

Standard SDLC

SDD Activities

Next iteration

Iterative spec update

Requirements

Gather and define needs

Write structured spec

Outcomes, scope, constraints, edge cases, verification criteria

Design

Architecture and tech decisions

Spec lives in the repo

API contracts, schemas, decisions recorded in versioned spec file
--- not a wiki or doc tool

Development

Write and review code

Spec-first AI coding

Agent reads spec as context

Ambiguity halts, not guesses

[NEEDS CLARIFICATION] markers

Testing

Verify against acceptance criteria

Spec-generated tests

Verification criteria -> test cases

Adversarial agent review

Separate verifier checks vs spec

Deployment

Release to production

Spec compliance gate

PR must cite or update the spec

Maintenance

Bug fixes, enhancements

Spec-anchored evolution

Update spec first, then code

SDD levels (adopt incrementally)

- Spec-first --- write spec, guide AI, spec may be discarded after
- Guardrail --- AI stops on ambiguity instead of guessing
- Spec-anchored --- spec stays alive as source of truth

Start with spec-first on 2-3 features. Graduate to spec-anchored for high-change areas. The spec is iterative --- write enough for the next cycle, not the entire feature upfront.